CHE 321 Self Study

Jasmin Bait

Janai Cedeño

Thao Nguyen

Michael Yanes

Introduction:

The class of multiphase reactors consist of three categories:

-Trickle bed reactor

-Fluidized bed reactor

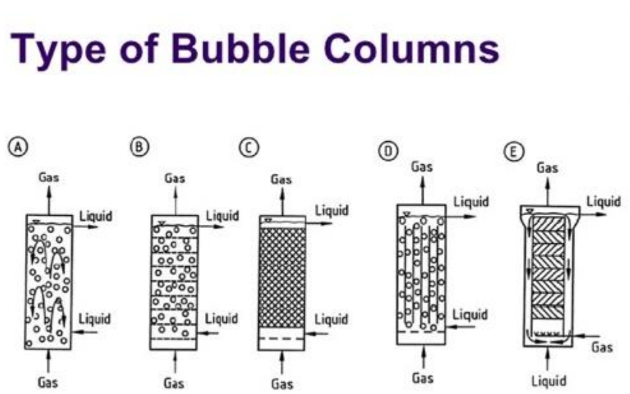
-Bubble column reactor

In this self-study project, we will only be discussing about Bubble Column Reactor.

What is Bubble Column Reactor (BCR)?

- BCR is a cylindrical vessel (with the aspect of height and diameter) filled with liquid, and a gas distributor at the bottom. The purpose is simply to mix the liquid phase. Gas mixed into the liquid by sparging, also known as gas flushing. This mixing technique requires less energy than mechanical stirring. Gas is entered from the bottom through a special designed distributor (commonly used distributors are metal micro porous spargers) with controlled pressure drop and nozzle sizes and distributed in the form of bubble into the liquid above. The bubbles rise at the velocity determined by the bubble size, the larger the bubbles the faster the rise. The sparger is designed so that it produces same size bubble. Therefore, most of the time, the bubbles rise with the same velocity. In the liquid, there may be some catalysts fluidized.

- There are several types of flow systems in BCR such as:
    - Homogeneous bubble flow: low gas velocity (<5cm/s) and uniform small sizes of bubbles
    - Heterogeneous flow: high gas velocity (> 5cm/s), different size of bubble, large diameter columns
    - Slug flow: high gas flow rates, large sizes of bubbles, small diameter columns



- BCR can be single or multi staged, batch or continuous phase. When there is a solid phase exist the reactor, it is usually described as a slurry BCR. Slurry BCR operating with a gas-liquid-solid system mostly under isothermal conditions. There is an electric heater installed in the column to maintain a constant temperature in the column.

- There are several type of bubble columns widely used in the industry:

  A. Simple Bubble Column
  B. Cascade Bubble Column with Sieve Tray
  C. Packed Bubble Column
  D. Multishaft Bubble Column
  E. Bubble Column with Static Mixers

Table 1
Biochemical applications of bubble column reactors

Bioproduct	Biocatalyst
Thienamycin	*Streptomyces cattleya*
Glucoamylase	*Aureobasidium pullulans*
Acetic acid	*Acetobacter aceti*
Monoclonal antibody	*Hybridoma cells*
Plant secondary metabolites	*Hyoscyamus muticus*
Taxol	*Taxus cuspidate*
Organic acids (acetic, butyric)	*Eubacterium limosum*
Low oxygen tolerance	*Arabidopsis thaliana*
Ethanol fermentation	*Saccharomyces cerevisiae*

- Application of BCR in Biochemical: BCR are widely used as multiphase contactors and reactors in chemical, biochemical and petrochemical industries. They have the ability to maintain high heat and mass transfer rates, compactness are low operating and maintenance costs.

Example Solution:

Bubble Colum Reactor

A bubble column is a liquid pool sparged by a process stream.

In the BCR, liquid can be in batch while the gas in continuous. Or both the liquid and gas phases can flow continuously. The gas superficial velocity is usually larger than the liquid superficial velocity, making the gas superficial velocity dominating. The gas superficial velocity, $U_{G,sup}$, usually varies up to 50 cm/s. Not very high but not a very low velocity.

$$U_{G,sup} \gg U_{L,sup}$$

And, the L/D ratio also varies from 2 to 20.

$$2 < \frac{L}{D} < 20$$

Gas liquid reactions are being carried out in the bubble column because of the better contact efficiencies. The gas and liquid flows in a vertical pipe. A distributor or a sparger can make a gas inlet for air to flow inside of the liquid.

Some advantages of using a BCR, is that it is a simple construction and has a low capital cost. There are no moving parts and has minimum maintenance for it. It can handle solids and is easy to control the temperature. A couple disadvantages may include that there is high pressure drop of the gas due to high static head of the liquid. And the G/L ratio area decreases if the ratio of height to diameter exceeds 12. This is due to bubble coalescence. Therefore, if the L/D ratio is greater than 12, then the gas to liquid area is being decreased and the bubble starts coalescing. So, if the height is very high, the bubble is sparging from the bottom, and because of the friction near the wall, the air will try to come in at the center of the column. There is a lower resistance in the middle of the column, therefore the bubbles will form larger bubbles in the middle of the column which will make the surface area reduced. That an issue with the BCR, so the bubble coalescence must be maintained.

There are different flow regimes used in BCRs. There is a bubbly flow which has small bubbles flowing through the column and it is usual calm. The throughput of the gas is usually small in bubbly flow and there is no bubble coalescence. The second flow regime is Churn-Turbulent Flow or Heterogeneous flow which operates at a higher velocity and has bubble coalescence that dominates in the BCR. The latter of the two is typically favored since the higher throughput system is wanted. The bubble size distribution is larger, and the mass transfer and the phenomena is more dominating in this system.

Gas Velocity (m/s) vs Column Diameter (m) and not the liquid velocity since it is very low or it can be zero, which is a batch liquid. Homogeneous regime or bubbly flow regime operates at a low velocity no matter what size the diameter of the gas inlet may be. And has small bubbles flowing upwards with small bubble coalescence. For homogeneous bubble flow regime, the gas volume fraction is less than around 5%. If the gas velocity is increased, the flow regime will transition to Churn-Turbulent regime. This will create bigger bubbles, hence bubble coalescence in the BCR. Now when the size of the bubble is approximately equal to the size (diameter) of the column, this is called Slug Flow Regime. This is typically for smaller diameter BCRs that has a diameter range of 4h to 10 centimeters. A diameter greater than a 10 to 15 cm will not happen in the BCR for slug flow regime. Typically, the Churn-Turbulent Regime is used to achieve higher yield, but it is a harder BCR to analyze since there is multiple bubble size distributions.

The modeling of the BCRs can be difficult, especially for the Churn-Turbulent Regime which has different bubble sizes flowing in the reactor. Some types of bubble columns include:

Single Stage, Mutli Stage, Mutli Channel, With Motionless Mixers, Loop Reactor, Jet Reactor, Downflow Bubble Column, Three-Phase Fluidized-Bed Reactor, and a Slurry Reactor.

Some bubble column design issues can be gas holdup. The gas holdup is directly related to the rise of velocity and the correlations of the form below are commonly used. ($u_{sg}^{a}$is the superficial velocity of the gases, ρ_l^b is liquid density, σ^c surface tension, and μ_l^d liquid viscosity. These four parameters).

$$\alpha \sim u_{sg}^{a} \rho_l^b \sigma^c \mu_l^d$$

The mass transfer coefficient is the k_la value. Correlations of the form shown below are commonly used. (now adding μ_g^e the viscosity of the medium, D^f diameter of the column and diameter of the bubble, and what is the buoyancy forces acting on it, $\Delta\rho^g$ delta rho)

$$k_l a \sim u_{sg}^{a} \rho_l^b \sigma^c \mu_l^d \, \mu_g^e D^f \Delta\rho^g$$

Want to increase the values for α and $k_l a$ since that will reduce the diameter of the bubble.

Also, the axial dispersion occurs in both the liquid and gas phase, and correlations for each are not available. The mixing time is also another design issue since the correlations are available for a limited number of systems. Liquid will move with the drag of the bubbles.

Then the volume, flow rates, and residence time also must be taken into consideration. And lastly, the type of flow regime used whether it be homogeneous flow, heterogeneous flow, or slug flow must be known for the design of the specific BCR. Typically, the heterogenous design is being used.

Other BCR design issues can include the accurate knowledge of the physical properties is important to know. Especially the effects of coalescence and mass transfer affecting chemicals.

Although good correlations are available for commonly studied air-water systems, these are limited to the ranges studied.

Correlations may not be available for large scale systems or systems with vessel geometries other than cylinders without internals.

Furthermore, experimental correlations may not accurately reflect changes in performance when flow regime transitions occur.

Determination of Bubble Size

Buoyancy force, gas momentum flux (this favors the detachment) surface tension force, drag force, and inertial force. These are the forces that determines the size of the bubble. Densities vary between the gas and liquid which affects the buoyancy. And the gas pressure inside of the bubble and the liquid pressure outside of the bubble must be taken into account, which is the pressure difference. This forces the bubble to move upward.

Buoyancy Force balance equation:

$$F_b = \frac{\pi d^3(\rho_l - \rho_g)g}{6} + \frac{\pi d_0^2}{4}(P_g - P_l)$$

(first term is the buoyancy term that includes d is diameter of the bubble, ρ_l is density of the liquid, ρ_g is density of the gas, and g is the gravitational forces) plus the orifice diameter squared d_0^2 times the pressure difference $(P_g - P_l)$ (last term is the pressure difference inside and outside of the bubble).

Gas momentum flux force of the bubble balance equation:

$$F_{mb} = \frac{\pi d_0^2}{4}\rho u_0^2$$

(u_0 is the orifice velocity and is equal to: $u_0 = \frac{Q}{Area*n} = \frac{4Q}{\pi d_0^2}$)

Surface tension force balance equation:

$$F_s = \pi d_0 \sigma$$

(σ is surface tension)

Drag force balance equation:

$$F_d = \frac{\pi}{4} d^2 c_p \frac{\rho_g u_b^2}{2}$$

(d is diameter of the bubble, bubble c_p area, ρ_g is density of the gas, and u_b is bubble rise velocity)

Inertial force balance equation:

$$F_i = (\rho_g V - \rho_l V_l)\frac{u_b}{t_d}$$

(ρ_g is density of gas times V the volume of the bubble minus ρ_l density of liquid V the volume of the liquid) all times u_b the bubble rise velocity divided by t_d the detached time

Amount of liquid moved around the bubble

Certain mass of the gas moving, virtual mass force (very high in the BCR) and at what rate the bubble is accelerating at.

What is the bubble acceleration?

This all determines the size of the bubble and surface tension also plays a role

But ultimately the most important terms are the buoyancy force and the surface tension force.

Simplest assumption is that all the other forces are negligible, besides the two previously mentioned. Then you will see this equation for the simplest case when buoyancy and surface tension is important.

$$\frac{\pi}{6}d^3(\rho_l - \rho_g)g = \pi d_0 \sigma$$

$$d = \left(\frac{6d_0\sigma}{(\rho_l - \rho_g)g}\right)^{\frac{1}{3}}$$

(d is the size of the bubble)

$$(\rho_l \gg \rho_g)$$

Forces that will help in detachment is equal

The force balance below can find help determine the bubble diameter in the BCR.

The force of buoyancy plus the force of momentum flux of gas is equal to the force of surface tension plus the drag force and the inertial force.

$$F_b + F_{mg} = F_S + F_D + F_i$$

The final equation is the transformation (velocity in terms of the flow rate):

We is Weber number which is important for determining the size of the bubble.

$$d = \left(\frac{6d_0\sigma}{(\rho_l - \rho_g)g}\right)\left(1 - \frac{We}{4}\right) + \left(\frac{81u_lQ_g}{\pi(\rho_l - \rho_g)gD}\right) + \left(\frac{135}{4\pi^2} + \frac{27\rho_g}{\pi^2\rho_l}\right)\left(\frac{\rho_lQ_g^2}{(\rho_l - \rho_g)gd^2}\right)$$

$$We = Weber\ number = \frac{inertial\ force}{surface\ tension\ force} = \frac{\rho_g u_0^2 d_0}{\sigma}$$

Safety Considerations:

When designing bubble column reactors some important factors to consider include the heat transfer which is a strong function of parameters in the reactor such as the superficial gas velocity, thermophysical properties of the liquid and solid particles, and the size and concentration of the particles. When designing a BCR one of the biggest safety concerns stems from undesirable multiplicity behavior developed from the existence of multiple steady states, these undesirable multiplicity behaviors may be observed depending on the parameters of the BCR. Specifically, the Stanton number for heat transfer and the Damkohler number are the most practically important parameters, with a higher Stanton number and a lower Damkohler number providing the better environment enhancing operability.

Resources

https://www.researchgate.net/publication/222662383_Bubble_Column_Reactors

https://link.springer.com/chapter/10.1007/978-94-009-4400-8_12

https://www.intechopen.com/books/computational-fluid-dynamics-basic-instruments-and-applications-in-science/cfd-for-the-design-and-optimization-of-slurry-bubble-column-reactors

http://web.ist.utl.pt/~ist11061/de/Equipamento/BubbleColumnReactors(review).pdf

https://www.youtube.com/watch?v=BSmLJmrq9r8

https://www.academia.edu/6834279/Multiplicity_and_sensitivity_analysis_of_Fischer_Tropsch_bubble_column_slurry_reactors_plug-flow_gas_and_well-mixed_slurry_model